Wenxin ZHANG

Contact

Kravis Hall 1190
665 West 130th St, Columbia University
New York, NY 10027

Phone: +1 (646) 726-3329
wzhang26@gsb.columbia.edu
https://wenxinzhang0.github.io/

Research Interests

My research focuses on dynamic resource allocation, especially for large-scale computing systems that power modern machine learning. I build application-grounded models and design simple, deployable algorithms with provable performance guarantees, drawing tools from stochastic modeling, optimization, and control theory.

Education

Columbia University, Graduate School of Business — New York, NY

- Ph.D. candidate in Decision, Risk, and Operations — 2021–2026 (expected)
- Advisors: Santiago R. Balseiro, Will Ma

Tsinghua University — Beijing, China

- B.E. in Industrial Engineering — 2017–2021
- Minor in Data Science and Technology

Journal Publications

J3. "Feature-Based Dynamic Matching," with Yilun Chen, Yash Kanoria, and Akshit Kumar. ***Operations Research***, forthcoming. [Link]

- **Third Place, Michael H. Rothkopf Junior Researcher Paper Prize Competition**, 2025 (Entrant: A. Kumar)
- **First Place, Jeff McGill RMP Best Student Paper Prize**, 2024 (Entrant: A. Kumar)

J2. "Dynamic Pricing for Reusable Resources: The Power of Two Prices," with Santiago R. Balseiro and Will Ma. ***Operations Research***, 2025. [Link]

J1. "A Gilmore-Gomory-Type Construction of Integer Programming Value Functions," with Seth Brown, Temitayo Ajayi, and Andrew J. Schaefer. ***Operations Research Letters***, 2021. [Link]

Refereed Conference Publications

C3. "Tail-Optimized Caching for LLM Inference," with Yueying Li, Ciamac Moallemi, and Tianyi Peng. **Neural Information Processing Systems (NeurIPS)**, 2025. [Link]

C2. "Distributed Load Balancing with Workload-Dependent Service Rates," with Santiago R. Balseiro, Robert Kleinberg, Vahab Mirrokni, Balasubramanian Sivan, and Bartek Wydrowski. **26th ACM Conference on Economics and Computation (EC)**, 2025. [Link]

C1. "Feature-Based Dynamic Matching," with Yilun Chen, Yash Kanoria, and Akshit Kumar. **24th ACM Conference on Economics and Computation (EC)**, 2023. [Link]

Working Papers

W4. "Distributed Load Balancing with Adversarial Arrivals" with Santiago R. Balseiro, Robert Kleinberg, and Balasubramanian Sivan.

W3. “Distributed Load Balancing with Workload-Dependent Service Rates” with Santiago R. Balseiro, Robert Kleinberg, Vahab Mirrokni, Balasubramanian Sivan, and Bartek Wydrowski. [Link]
- Under Review, *Operations Research*.
- **Finalist, Applied Probability Society Best Student Paper Competition, 2025**

W2. “Tail-Optimized Caching for LLM Inference” with Yueying Li, Ciamac Moallemi, and Tianyi Peng. [Link]
- Accepted at 3rd ICML Workshop on Efficient Systems for Foundation Models, 2025.
- Accepted at ML x OR Workshop, NeurIPS, 2025.

W1. “Leveraging Offline Data for Online Decision-Making in Bayesian Multi-Armed Bandits,” with Santiago R. Balseiro and Will Ma.

Experiences

Student Researcher, Google Research NYC — Summer 2024 & 2025
- Hosted by Balasubramanian Sivan. Worked on distributed load balancing.

Data Science Intern, ParkHub — Fall 2023
- Designed algorithms for real-time pricing using historical data and booking trends.

UGVR Research Assistant, Stanford University — Summer 2020

Visiting Undergraduate Researcher, Rice University — Summer 2019

Honors and Awards

Paul and Sandra Montrone Doctoral Fellowship	2025–2026
Finalist, Applied Probability Society Best Student Paper Competition	2025
Deming Doctoral Fellowship	2025–2026
Doctoral Fellowship, Columbia Business School	2021–2026
Tsinghua University Outstanding Undergraduate Student Award	2021
Beijing Outstanding Undergraduate Student Award	2021
National Scholarship, Ministry of Education of China	2020
Baosteel Scholarship, Baosteel Education Fund	2020

Teaching Experience

Columbia University, Teaching Assistant
- Operations Management (EMBA Core) — Fall, Spring 2024
- Revenue and Supply Chain Management (MBA) — Summer 2024
- Managerial Statistics (EMBA Core) — Fall 2023, 2022
- Managerial Statistics (MBA Core) — Fall 2023
- Healthcare Management, Design, and Strategy (MBA) — Fall 2022, 2021

Selected Talks

Distributed Load Balancing with Workload-Dependent Service Rates
- INFORMS Annual Meeting, Atlanta — October 2025
- Cornell Young Researchers Workshop, Ithaca — October 2025
- The Chinese University of Hong Kong, Shenzhen, School of Data Science, Shenzhen — September 2025
- Next-Gen Scholar’s Symposium, Singapore — September 2025
- Third Purdue Operations Conference, West Lafayette — August 2025
- Economics and Computation (EC) conference, Stanford — July 2025
- INFORMS Applied Probability Society Conference (APS), Atlanta — July 2025
- Marketplace Innovation Workshop (MIW), virtual — May 2025

- NYC Operations Day PhD Colloquium, New York — April 2025

Leveraging Offline Data for Online Decision-Making in Bayesian Multi-Armed Bandits
- INFORMS Annual Meeting, Seattle — October 2024

Dynamic Pricing for Reusable Resources: The Power of Two Prices
- POMS-HK International Conference, Hong Kong — Jan 2025
- INFORMS Annual Meeting, Phoenix — Oct 2023
- Revenue Management and Pricing (RMP) conference, London — July 2023
- Manufacturing & Service Operations Management Conference (MSOM), Montreal — Jun 2023
- Marketplace Innovation Workshop (MIW), virtual — May 2023
- NYC Operations Day PhD Colloquium — May 2023

Professional Services

Co-organizer for the NYC Operations Day PhD Colloquium — 2023
Student social co-chair for the Decision, Risk, and Operations division — 2022-2024

References

Prof. Santiago R. Balseiro
George E. Warren Professor of Business
Decision, Risk, and Operations
Columbia Business School
E-mail: srb2155@gsb.columbia.edu
Phone: +1 (646) 460-1264

Prof. Will Ma
Roderick H. Cushman Associate Professor
Decision, Risk, and Operations
Columbia Business School
E-mail: wm2428@gsb.columbia.edu
Phone: +1 (617) 938-9151

Prof. Tianyi Peng
Assistant Professor
Decision, Risk, and Operations
Columbia Business School
E-mail: tianyi.peng@columbia.edu
Phone: +1 (617) 230-1062

Prof. Bobby Kleinberg
Professor of Computer Science
Cornell University
E-mail: rdk@cs.cornell.edu
Phone: +1 (607) 255-9200

Dr. Balu Sivan
Research Scientist
Google New York
E-mail: balusivan@google.com
Phone: +1 (212) 565-7760

Last updated on November 17, 2025